



VC9000E Series Video Encoder FPGA Verification Guide

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This document is compatible with
VC9000E series (VC9000E, VC9000LE, VC9000EJ) hardware versions 1.x

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Preface

This preface introduces the abbreviations and conventions used in this document and provides related documents for your reference.

Abbreviations

Abbreviations	Full name
API	Application Programming Interface
PC	personal computer
PCI-e	Peripheral Component Interconnect Express
VPU	Video Processing Unit

Conventions

- Hexadecimal numbers are indicated by the prefix "0x".
For example, 0x32CF.
- Code snippets and file paths are given in Consolas font.

Reference

- Hantro VC9000E Series Video Encoder Hardware Integration Guide* is the primary integration reference manual for this IP.
- Hantro VC9000 Series Video Encoder Hardware Simulation Guide* describes the simulation environment including the test bench and simulation scripts for the IP core.
- Hantro VC9000E V1x Video Encoder TestBench* and *Hantro VC9000E V1x JPEG Encoder TestBench* describe the usage of the command line API of the Hantro VC9000E HEVC/AVC VPU.

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1 Overview

This document guides you through the FPGA verification process of VC9000E series video encoder on either an ARM platform or a PC platform with a PCI-e bus. The process includes the following steps in order:

- [Section 2, Installing Drivers in Linux](#)
- [Section 3, Verifying FPGA Against C-Model](#)

1.1 Introduction to VC9000E Series Video Coder

Designed for easy integration onto the SoC, Hantro VC9000E series video decoder is a single-core solution based on silicon-proven designs deployed in millions of smartphones, feature phones, digital still cameras, tablets, set-top boxes, surveillance cameras, and video cameras worldwide.

The multi-format VC9000E series video decoder provides up to 4K@60fps throughput decoding for both HEVC and H.264. It is one of the smallest HEVC + H.264 video encoder IP solutions and features low power consumption and negligible load on the host CPU. The integrated reference frame compression (RFC) provides significant savings on system bandwidth, and extensive use of multi-level hierarchical clock gating minimizes the dynamic power consumption.

Embedded within the VC9000E series video decoder is VeriSilicon Universal Compression IP, DEC400 (Decoder/Encoder Core), a versatile and efficient compression module which, when multi-instantiated, can provide a unified, lossless pixel compression and decompression infrastructure for the SoC.

1.2 Directory and File Aliases

This guide uses aliases listed in [Table 1](#) to represent directories or files frequently used during verification.

Table 1. Directory and File Aliases

Alias	Directory/File
CODE_BASE	The root directory of VC9000E series release package
SOFTWARE_BASE	CODE_BASE/software
SYSTEM_BASE	CODE_BASE/system
TESTBENCH_BASE	CODE_BASE/software/linux_reference/test
TEST_BINARY	An executable software-generated binary file that resides in the TESTBENCH_BASE subdirectory.

2 Installing Drivers in Linux

To facilitate FPGA verification, you need to install the following drivers in Linux:

- **memalloc**
A kernel driver that manages memory for the VC9000E series device encoder in test environment. It divides the memory into chunks and assigns base addresses to these chunks.
- **VC9000E series device driver**
A kernel driver that interacts with hardware.

2.1 Prerequisites

Before installing the memalloc and VC9000E series device drivers, make sure that a VC9000E series release package is available.

2.2 Installing memalloc

1. Open the file `SOFTWARE_BASE/linux_reference/memalloc/Makefile`.
2. In the Makefile file, specify the memory that is used only by the VC9000E series video encoder by setting the following macros:
 - **HLINA_START**: the base address of the memory
 - **HLINA_SIZE**: the size of the memory in MB

Make sure that the specified memory is not managed by the Linux system and is for the encoder use only.

Example:

```
HLINA_START := 0x04000000
HLINA_SIZE := 250
```

3. (Required only for ARM) Modify the settings of the following macros:
 - Set **KDIR** to the path of your kernel directory.
 - Set **CROSS_COMPILE** to your toolchain path.
 - Set **ARM_CROSS_COMPILE** to y.
4. In the shell terminal, navigate to the directory `SOFTWARE_BASE/linux_reference/memalloc`.
5. Build the source code of the memalloc by using the make command.
After the build, a file named `memalloc.ko` for the driver is generated in the `memalloc` directory.
6. Run the `memalloc_load.sh` shell script to install the memalloc.
After the script is executed, `memalloc.ko` is inserted onto the Linux platform.

2.3 Installing VC9000E Series Device Driver

1. (Required only for ARM) Edit the file `SOFTWARE_BASE/linux_reference/kernel_module/linux/Makefile` as follows:
 - Set `KDIR` to the path of your kernel directory.
 - Set `CROSS_COMPILE` to your toolchain path.
 - Set `ARM_CROSS_COMPILE` to `y`.
2. In the shell terminal, navigate to the directory `SOFTWARE_BASE/linux_reference/kernel_module/linux`.
3. Build the source code of the VC9000E series device driver by using the `make` command.
After the build, a file named `vsi_vcx.ko` for the driver is generated in the `kernel_module` directory.
4. Run the `driver_load.sh` shell script to install the VC9000E series device driver.
After the script is executed, `vsi_vcx.ko` is inserted onto the Linux platform.

3 Verifying FPGA Against C-Model

Verification uses bit streams generated by FPGA test binaries to compare with the bit stream generated by the C-Model binary.

Prerequisites

Before you build and run test binaries, make sure:

- memalloc and VC9000E series device drivers are installed.
For information about how to install the drivers, see [Chapter 2, Drivers in Linux](#).
- The hardware simulation version matches the hardware RTL version.
- C-Model compiling and FPGA compiling use the same `Ctrl_sw`.
For example, the `SOFTWARE_BASE/source/` directory is used for both C-Model and FPGA.
Note: The `Ctrl_sw` contains the frame rate control software.
For more information about the C-Model, see *Hantro VC9000E V1x Video Encoder TestBench* and *Hantro VC9000E V1x JPEG Encoder TestBench*.
- (Required only for ARM) ARM cross compiler and ARM architecture options are properly set in `SOFTWARE_BASE/build/globalrules`.

Procedure

1. In the shell terminal, navigate to the directory for test binary building.

The directory varies among different stream formats, as listed in [Table 2](#).

Table 2. Directories for Test Binary Building

Stream Format	Directory for Building
HEVC	TESTBENCH_BASE/hevc
H.264	TESTBENCH_BASE/h264
JPEG	TESTBENCH_BASE/jpeg

2. Build the test binary.
 - If you are using the ARM platform, use the `make libclean clean armlinux` command.
 - If you are using the PC platform with PCI-e, use the `make libclean clean pci` command.

After the test binary is build, a `TEST_BINARY` file is generated. The location of the generated file varies as listed in [Table 3](#).

Table 3. TEST_BINARY File Locations

Stream Format	TEST_BINARY File Location
HEVC	TESTBENCH_BASE/hevc/hevc_testenc
H.264	TESTBENCH_BASE/h264/h264_testenc
JPEG	TESTBENCH_BASE/jpeg/jpeg_testenc

- Use the C-Model binary to run cases.

Each case generates a bit stream that is regarded as golden data.

- Run the TEST_BINARY file with the same options used in Step 3.

To list all command options in the shell terminal, specify -H in the command as shown in the following table.

Table 4. Commands for Listing Command Options

Standard	Command
HEVC	hevc_testenc -H
H.264	h264_testenc -H
JPEG	jpeg_testenc -H

If the encoded stream of each case generated by FPGA is bit exact with the one generated by C-Model, FPGA verification is passed.

Document Revision History

This section describes top level differences in the versions of this document.

Note: This document is not necessarily updated for each patch or minor revision.

Document Revision	Date	Compatible cores	Description
1.10	2022-11-02	VC9000E series	• Chapter 2, Installing Drivers in Linux ▪ Updated the procedure and directories for building memalloc and VC9000E series device driver. ▪ Added the macro for setting cross-compile toolchain to the path of your kernel directory. • Chapter 3, Verifying FPGA Against C-Model Updated the commands and directories for building test binary. • Refined the document content and structure.
1.02	2022-09-01	VC9000E series	Globally changed VC9000E to VC9000E series to be compatible with more VC9000E series products.
1.01	2022-02-03	VC9000E series	Legal Notices, General: update branding layout to include VeriSilicon.
1.00	2021-03-15	VC9000E series	Initial.